

Лабораторная работа №5

Менеджер паролей pass и управление файлами конфигурации

Ван Хаоцзянь

РУДН

2026-03-15

Цель работы

- Изучение менеджера паролей pass
- Получение навыков работы с GPG-ключами
- Освоение семантической структуры базы паролей
- Настройка синхронизации паролей через Git
- Изучение системы управления конфигурациями chezmoi
- Освоение работы с шаблонами

Установка ПО

- **Установка pass:** `sudo apt install pass pass-otp -y`
- **Установка chezmoi:**

```
sh -c "$(wget -qO- get.chezmoi.io)" -- -b ~/bin
echo 'export PATH=$PATH:$HOME/bin' >> ~/.bashrc
```

```

[base] ~$ sudo apt-get install libqrencode4 -y
Fetched 3540 kB in 2s (1452 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libqrencode4 can be upgraded. Run 'apt list --upgradable' to see them.
(base) [base] ~$ cd /root/.ssh/ && run/study/2026-2027/Операционные системы/os-intro/labs/lab05$ sudo apt install pass -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  dkms libdecor-0 libdecor-0-plugin-1-gtk libgsoap-2.8.132t64 liblzf1 libqt5help5 libqt5opengl5t64 libqt5sprintsupport5t64 libqt5sql5-sqlite libqt5sql5t
  libsamplerate0 libstdl2-2.0-0 libtpm20 libvncserver1 linux-headers-6.8.0-101 linux-headers-6.8.0-101-generic linux-headers-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libqrencode4 qrencode wl-clipboard xclip
Suggested packages:
  libxml-simple-perl python
The following NEW packages will be installed:
  libqrencode4 pass qrencode wl-clipboard xclip
0 upgraded, 5 newly installed, 0 to remove and 118 not upgraded.
Need to get 137 kB of archives.
After this operation, 478 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 libqrencode4 amd64 4.1.1-1build2 [25.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 pass all 1.7.4-6 [34.5 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 qrencode amd64 4.1.1-1build2 [26.1 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 wl-clipboard amd64 2.2.1-1build1 [33.4 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 xclip amd64 0.13-3 [17.6 kB]
Fetched 137 kB in 0s (573 kB/s)
Selecting previously unselected package libqrencode4:amd64.
(Reading database ... 386382 files and directories currently installed.)
Preparing to unpack .../libqrencode4_4.1.1-1build2_amd64.deb ...
Unpacking libqrencode4:amd64 (4.1.1-1build2) ...
Selecting previously unselected package pass.
Preparing to unpack .../archives/pass_1.7.4-6_all.deb ...
Unpacking pass (1.7.4-6) ...
Selecting previously unselected package qrencode.
Preparing to unpack .../qrencode_4.1.1-1build2_amd64.deb ...

```

Подготовка GPG-ключа

- **Проверка наличия ключей:** `gpg --list-secret-keys`
- **Создание ключа:** `gpg --full-generate-key`

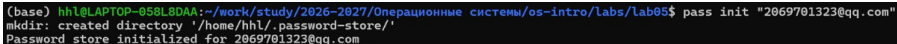
```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ gpg --list-secret-keys --keyid-format LONG
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
/home/hhl/.gnupg/pubring.kbx
-----
sec   rsa4096/83F0DBF2841BFA82 2026-03-09 [SC]
      52DF1A31909833E096A2FBD083F0DBF2841BFA82
uid           [ultimate] sunshengjie <2069701323@qq.com>
ssb   rsa4096/2B27EF74E37A7DC7 2026-03-09 [E]
```

Рисунок 2: Создание GPG ключа

Инициализация pass

- **Инициализация хранилища:**

```
pass init ""
```



A terminal window showing the execution of the 'pass init' command. The prompt is '(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05\$'. The command entered is 'pass init "2069701323@qq.com"'. The output shows 'mkdir: created directory '/home/hhl/.password-store/'' and 'Password store initialized for 2069701323@qq.com'.

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ pass init "2069701323@qq.com"
mkdir: created directory '/home/hhl/.password-store/'
Password store initialized for 2069701323@qq.com
```

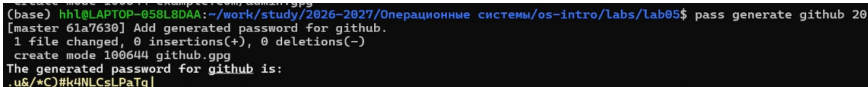
Рисунок 3: Инициализация pass

- **Настройка Git синхронизации:**

```
pass git init
pass git remote add origin git@github.com:haoczan2/pa
```

- **Добавление паролей:**

```
pass insert example.com  
pass insert work/company-email
```



```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ pass generate github 20  
[master 61a7630] Add generated password for github.  
1 file changed, 0 insertions(+), 0 deletions(-)  
create mode 100644 github.gpg  
The generated password for github is:  
.u&(*C)#k4NLCsLPaTq|
```

Рисунок 4: Добавление паролей

- **Генерация пароля:** `pass generate github 20`
- **Просмотр паролей:** `pass`
- **Копирование в буфер:** `pass -c example.com`

Настройка browserpass

• Установка native хоста:

```
sudo apt install webext-browserpass -y
```

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05/browserpass$ sudo apt install webext-browserpass -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  dkms libdecor-0-0 libdecor-0-plugin-1-gtk libgsoap-2.8.132t64 liblzf1 libqt5help5 libqt5opengl5t64 libqt5sprintsupport5t64 libqt5sql5-sqlite libqt5sql5t
  libssamplerate0 libstdl2-2.0-0 libtptm0 libvncserver1 linux-headers-6.8.0-101 linux-headers-6.8.0-101-generic linux-headers-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  firefox
The following NEW packages will be installed:
  firefox webext-browserpass
0 upgraded, 2 newly installed, 0 to remove and 118 not upgraded.
Need to get 1634 kB of archives.
After this operation, 5956 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 firefox amd64 1:1snap1-0ubuntu5 [77.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 webext-browserpass amd64 3.7.2-1ubuntu0.24.04.3 [1557 kB]
Fetched 1634 kB in 2s (860 kB/s)
Preconfiguring packages ...
Selecting previously unselected package firefox.
(Reading database ... 386472 files and directories currently installed.)
Preparing to unpack .../firefox_1%3alsnap1-0ubuntu5_amd64.deb ...
==> Installing the firefox snap
==> Checking connectivity with the snap store
==> Installing the firefox snap
Download snap "core22" (2339) from channel "stable" 43% 3.79MB/s 11.6s
progress [ 1%] [#####]
```

Рисунок 5: Установка browserpass

• Установка расширения в браузере Chrome

- **Проверка установки:** `chezmoi --version`

```
(base) hh1@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05/browserpass$ chezmoi --version
chezmoi version v2.69.4, commit c4c669c9f2f329233a85802014d26fba3c58a4a4, built at 2026-02-11T08:59:37Z, built by goreleaser
```

Рисунок 6: Версия chezmoi

- **Инициализация с репозиторием:**

```
chezmoi init git@github.com:haoczan2/dotfiles.git
```


Работа с chezmoi

- **Применение изменений:** `chezmoi apply -v`
- **Просмотр переменных:** `chezmoi data`

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05$ chezmoi data
{
  "chezmoi": {
    "arch": "amd64",
    "args": [
      "chezmoi",
      "data"
    ],
    "cacheDir": "/home/hhl/.cache/chezmoi",
    "command": "data",
    "commandDir": "/home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab05",
    "config": {
      "add": {
        "encrypt": false,
        "secrets": "warning",
        "templateSymlinks": false
      },
      "age": {
        "args": null,
        "command": "age",
        "identities": null,
        "identity": "",
        "passphrase": false,
        "recipient": "",
        "recipients": null,
        "recipientsFile": "",
        "recipientsFiles": null,
        "suffix": ".age",
        "symmetric": false,

```

Рисунок 7: Переменные шаблонов

- **Тестовый шаблон:**

```
echo 'Имя хоста: {{ .chezmoi.hostname }}' > test.tpl  
chezmoi execute-template < test.tpl
```

```
(base) bb1@ASTOR-B581-B0A6:~/work/study/2026-2027/Ознакомление_системы/os-intro/labs/lab05/test_templates$ chezmoi execute-template < test-template.tpl > test-template-output.txt
```

Рисунок 8: Тестирование шаблона

- **Добавление файла как шаблона:**

```
chezmoi add --template ~/.bashrc
```

Ежедневные операции

- **Обновление:** `chezmoi update -v`
- **Редактирование:** `chezmoi edit ~/.bashrc --apply`
- **Статус:** `chezmoi status`
- **Просмотр изменений:** `chezmoi diff`

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab05/browserpass$ chezmoi status
```

Рисунок 9: Статус chezmoi

- Установлен и настроен менеджер паролей pass
- Создан GPG-ключ для шифрования паролей
- Настроена синхронизация паролей через Git
- Установлено браузерное расширение browserpass
- Изучена система управления конфигурациями chezmoi
- Освоено создание и тестирование шаблонов
- Выполнены все операции по управлению паролями и конфигурациями Спасибо за внимание!